

Stochastic Mobility Integration into Residential Energy Hubs

ESARS-ITEC24 Paper ID 123

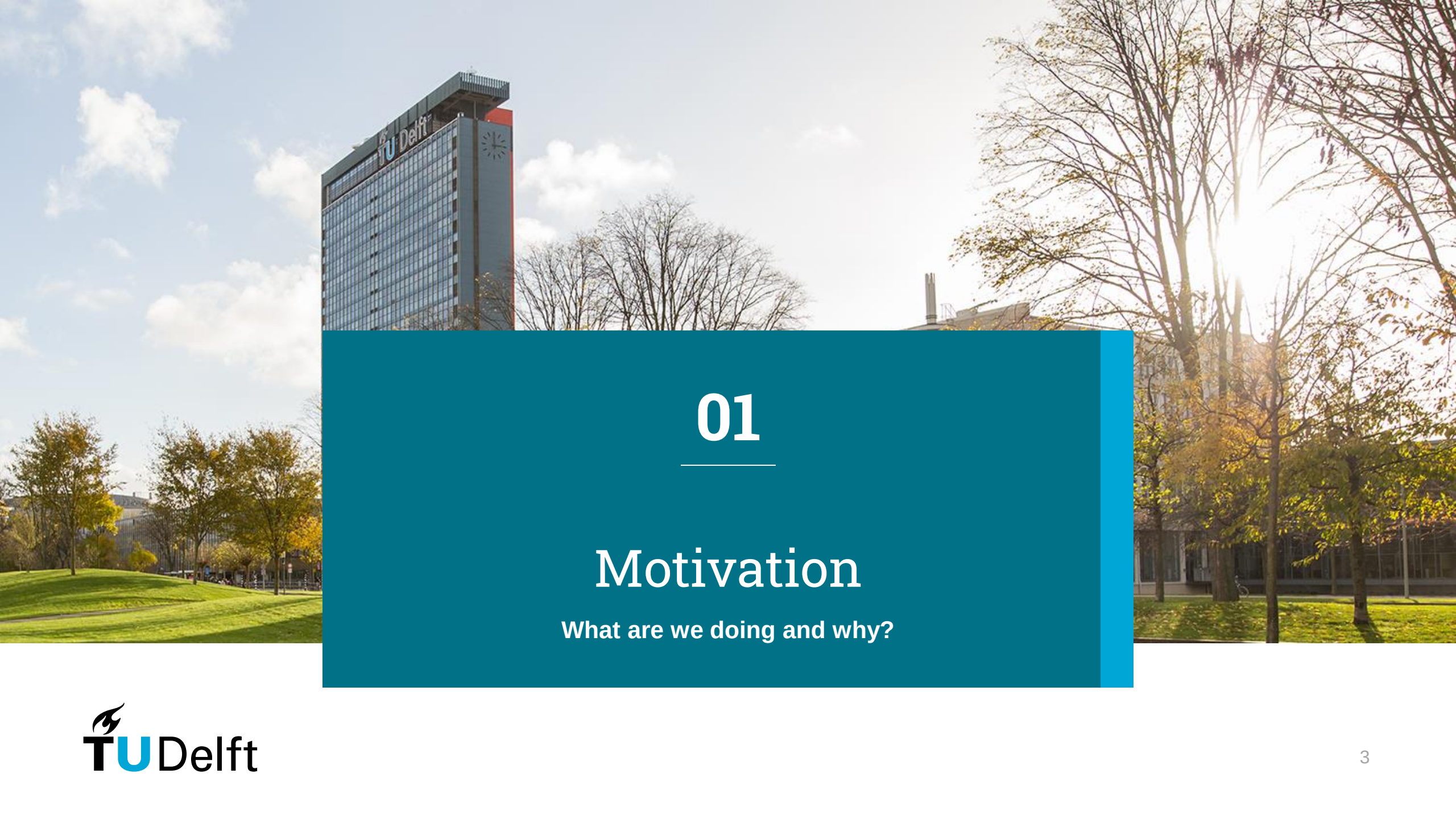
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27/11/2024

Content

1. Motivation
 2. Modelling & Algorithms
 3. Case study
 4. Conclusions
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01

Motivation

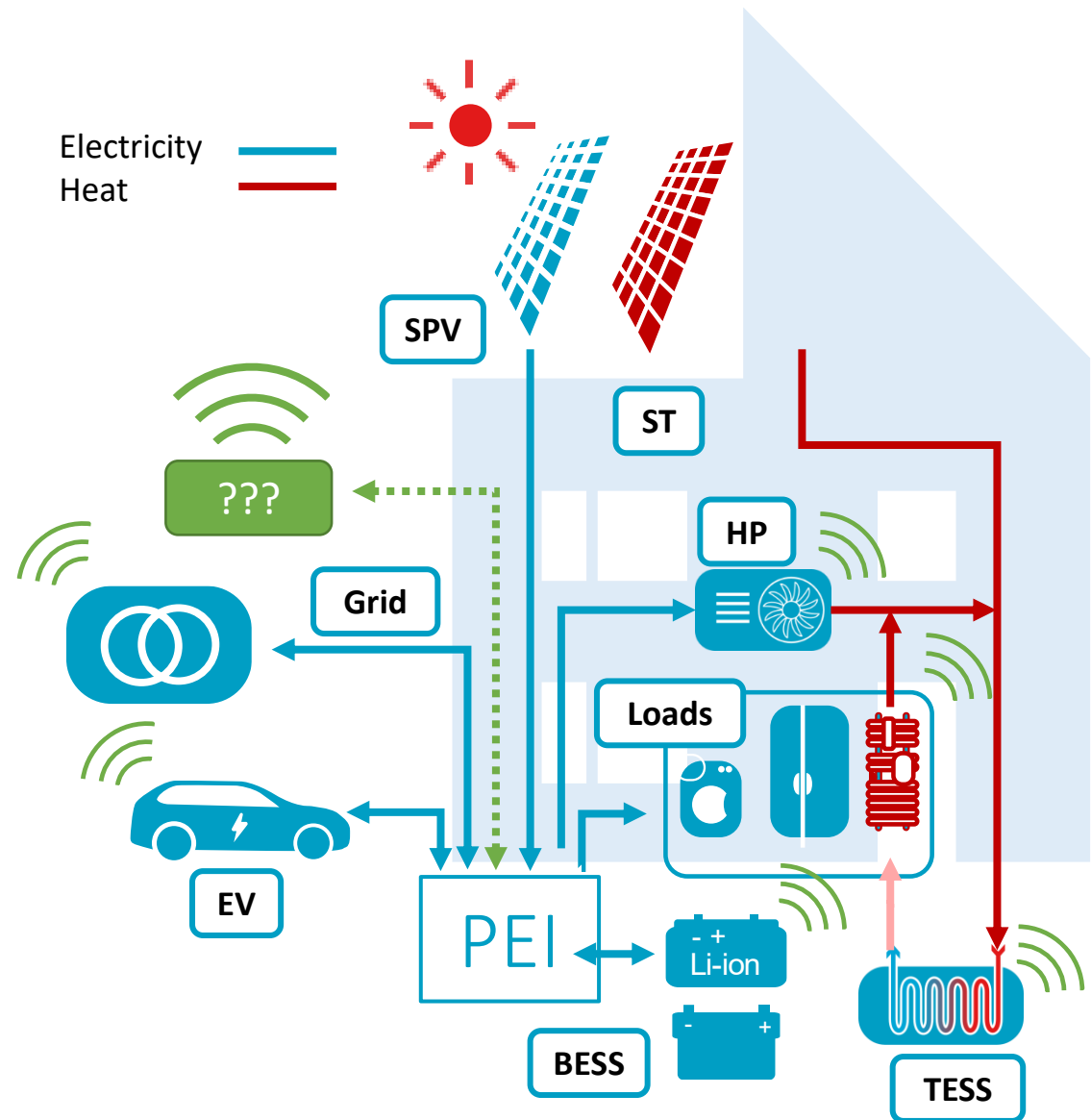
What are we doing and why?

Motivation

Electrified building

- A lot of devices → complexity
 - Thermal
 - Electrical
- Diverse uncertainty → robustness
- Dynamic environment → speed

Are all of the devices necessary?
What building service is supplied
by which device?



State of the Art

- Stochastic setup w/Random Field Opt. Ref. [Pulsipher \(2022\)](#)
- EV departure/arrival has a defined distribution.

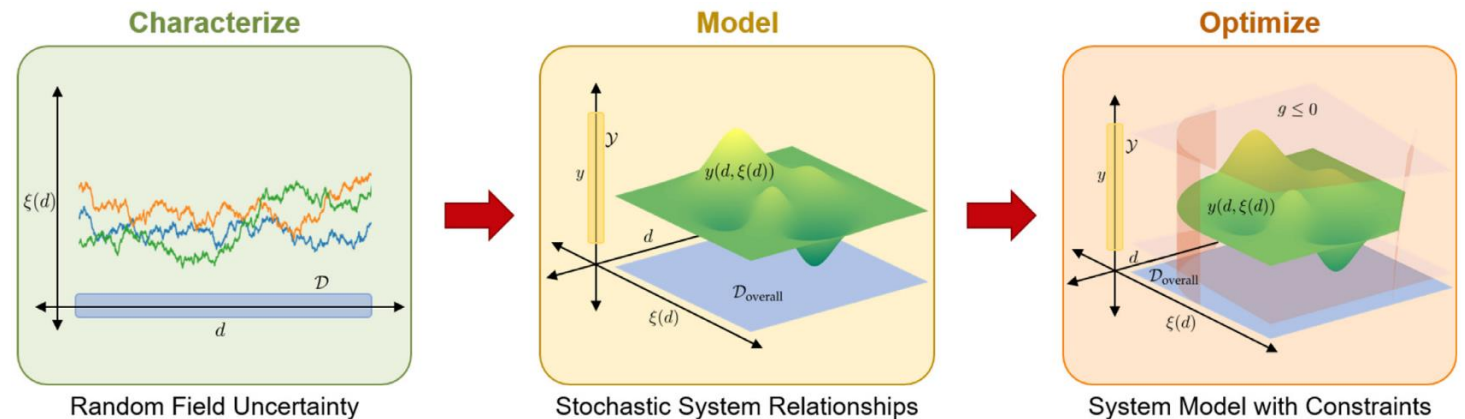
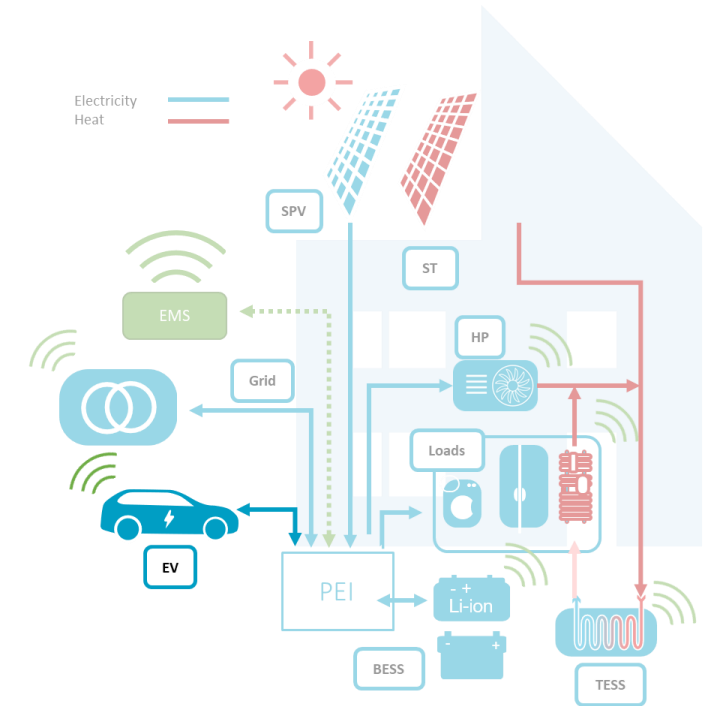
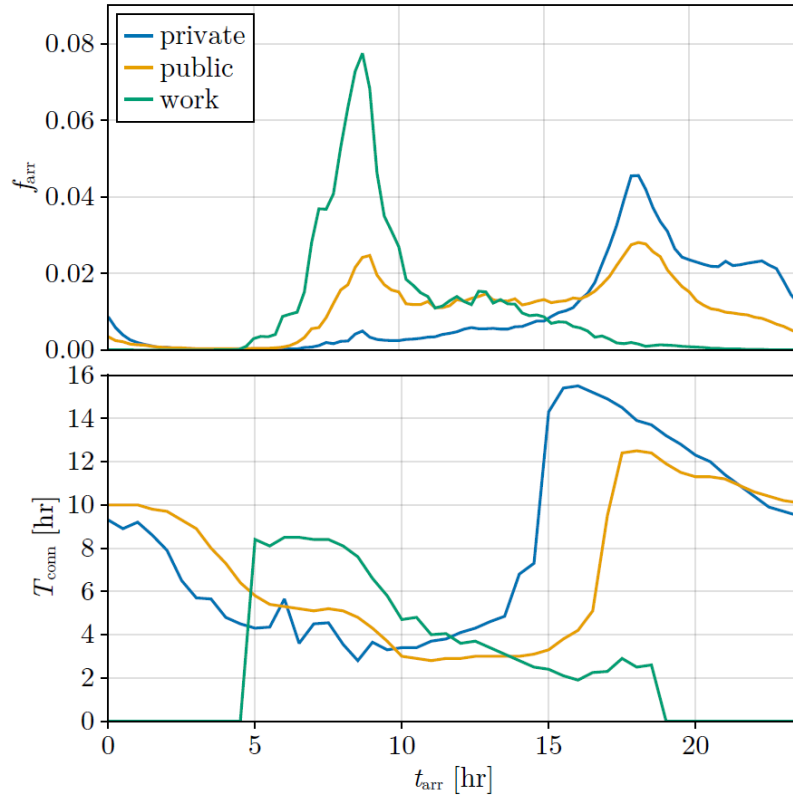


Fig. 1. Summary of proposed random field optimization framework.



02

Modelling & Algorithms

How do we start?

2. Modelling

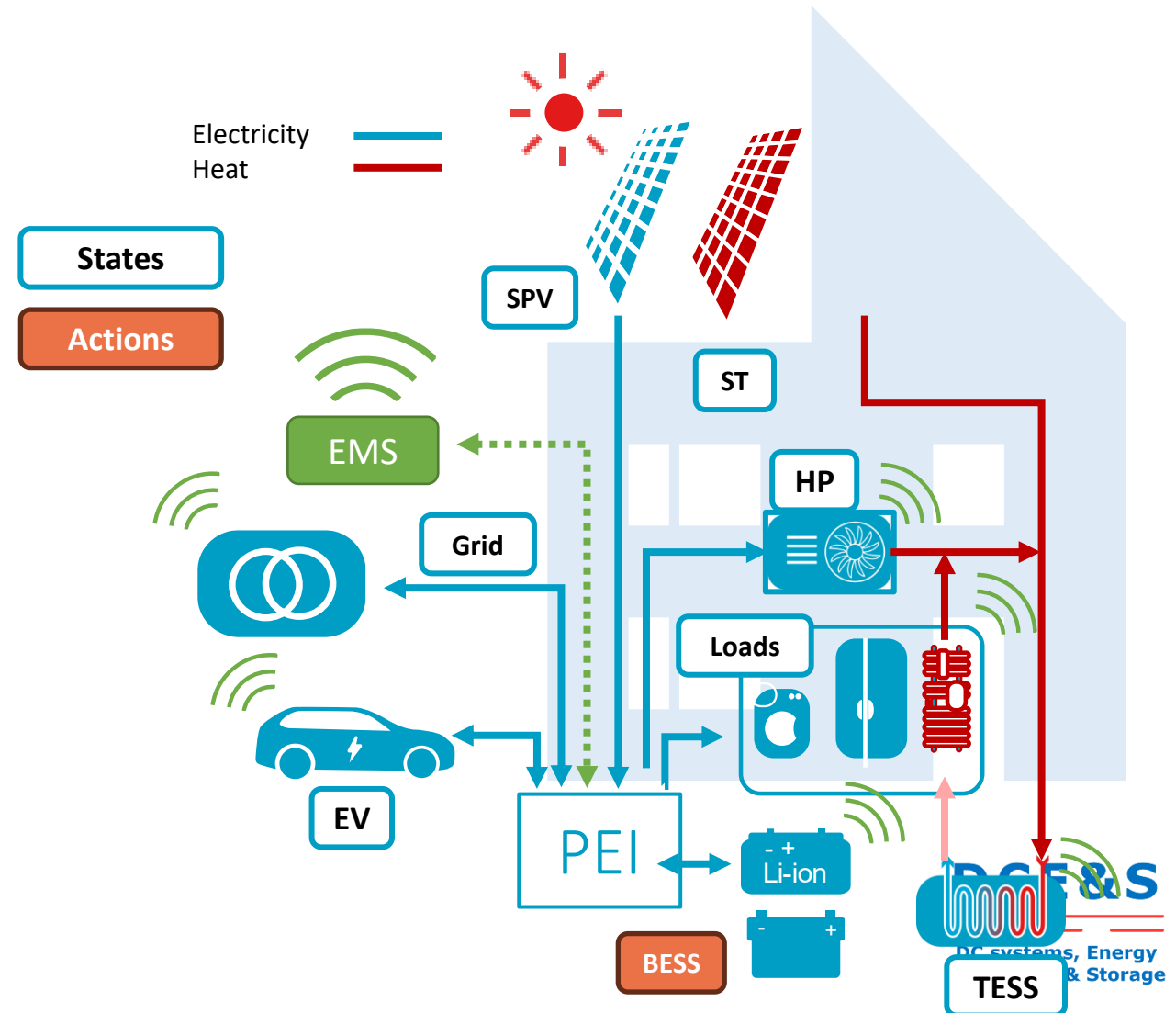
BESS Sizing Problem

$$\begin{aligned} \min_{\pi} \quad & \mathbb{E}_{t_{arr}, n_{EV}} [C_i] + c_{BESS} \cdot Q_{BESS} \\ \text{s.t.} \quad & S_{a,i,t+1} = S_{a,t}^M(S_{a,i,t}, Q_{BESS}, W_{i,t+1}) \\ & Q_{BESS}^* = X_t^\pi(S_{a,i,t}) \in \mathcal{Q} \\ & S_{a,i,t} \in \mathcal{S} \end{aligned}$$

With

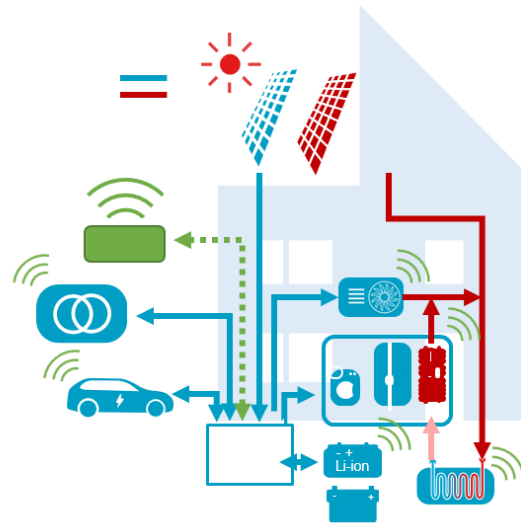
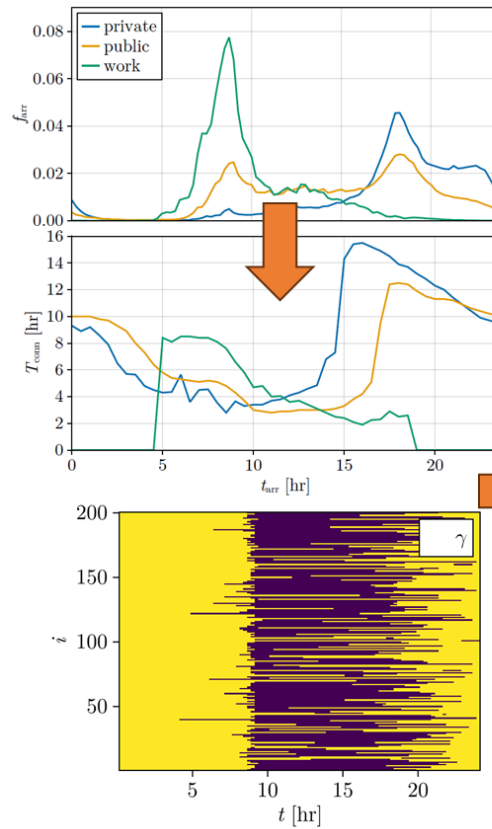
$$\begin{aligned} \forall a \in \mathbb{A}, i \in [1, N_s] \\ a = \{\text{SPV, grid, } n_{EV}, \text{BESS, HP, ST, TESS}\} \\ C_i = C_{g,i} + p_{\text{SoCDep},i} \\ C_{\text{grid},i} = w_{\text{grid}} \sum_0 \lambda_t \cdot P_{\text{grid},i,t} \cdot \Delta t \\ p_{\text{SoCDep},i} = w_{\text{SoC}} \left\| \varepsilon_{\text{SoC},i} \right\|_2^2 \\ \varepsilon_{\text{SoC},i} = \text{SoC}_{n_{EV},i}(t_{\text{dep},i}) - \text{SoC}_{n_{EV},\text{dep}} \\ \forall n_{EV} = 1, \dots, N_{EV} \end{aligned}$$

$S_{a,t}$ is the state of the system
 Q_{BESS}^* is the optimal decision
 W_{t+1} is exogenous information



2. Modelling - Workflow

Availability trajectories
Random Field sampling
 $t_{arr,n_{EV}} \sim \mathcal{T}_{arr}$

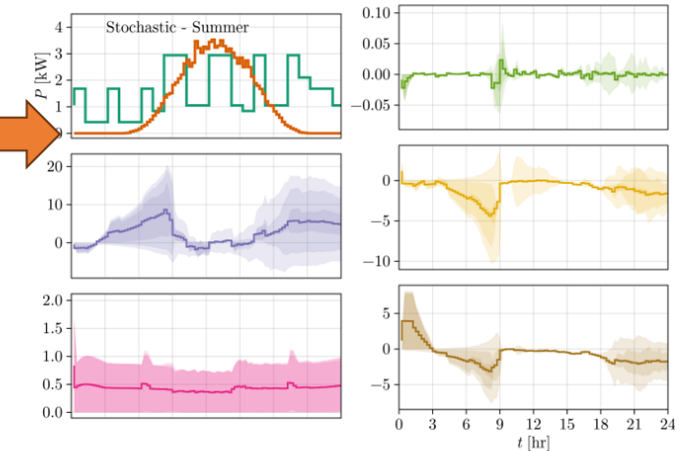


Planning & Sizing
Random Field Optimization
$$\min_{Q_{BESS}^* \in \mathcal{Q}} \mathbb{E}_{t_{arr,n_{EV}}} [C_i] + c_{BESS} \cdot Q_{BESS}$$

s. t. $S_{a,i,t+1} = S_{a,i,t}^M(S_{a,i,t}, Q_{BESS}, W_{i,t+1})$
 $Q_{BESS}^* = X^\pi(S_{a,i,t}) \in \mathcal{Q}$

Planner	N_s	# of Opt
Det	N_s	N_s
Det. Mean	1	1
RFO	N_s	1

Stochastic Dispatch
Optimal BESS size





03

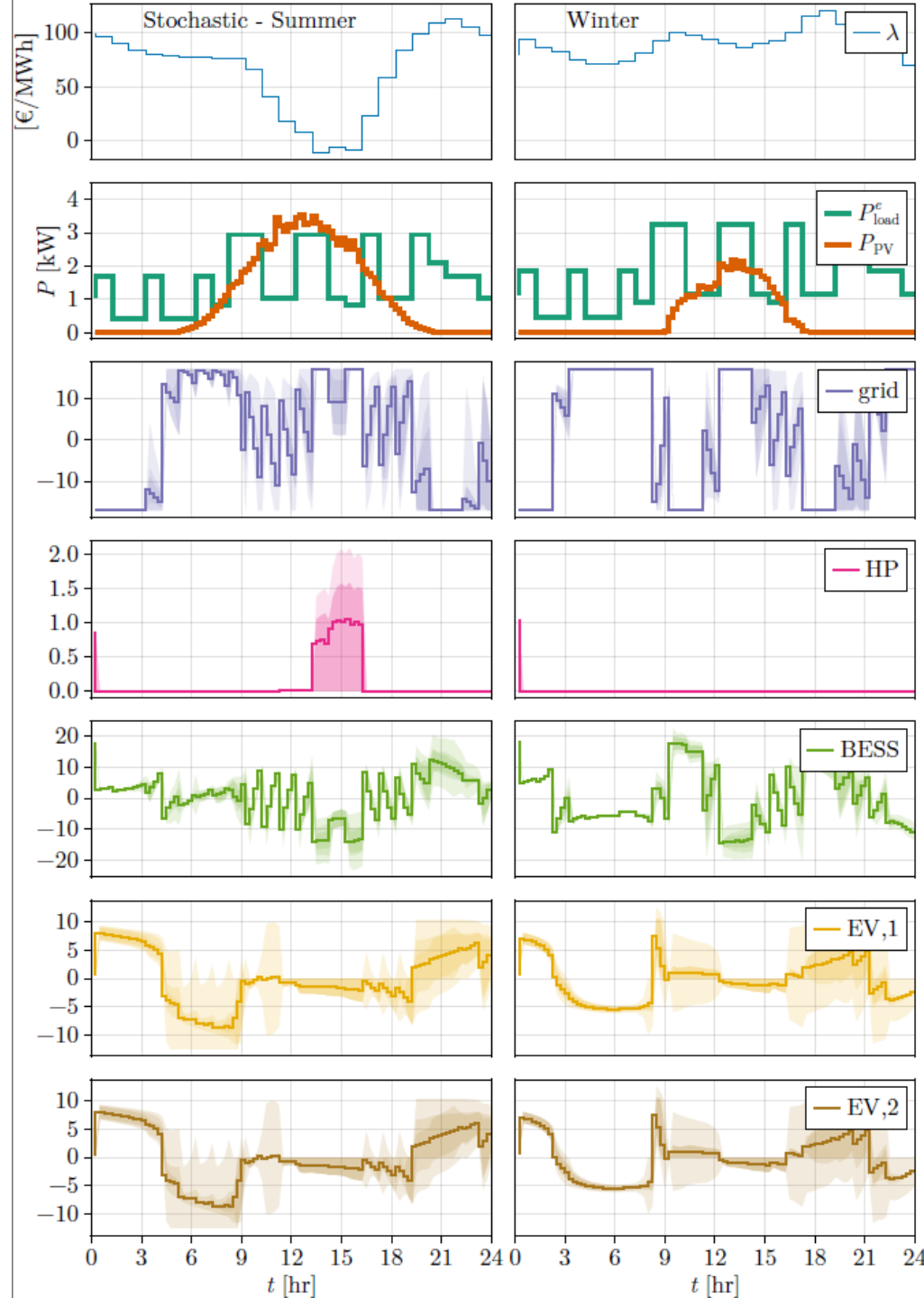
Case study

What is the real impact of our actions?

3. Case Study I

- The optimal BESS capacity is chosen, and its corresponding dispatch.
- Deterministic setup. → each sample is opt within itself
- RFO setup. → all samples are co-optimized leading to lower mean cost.

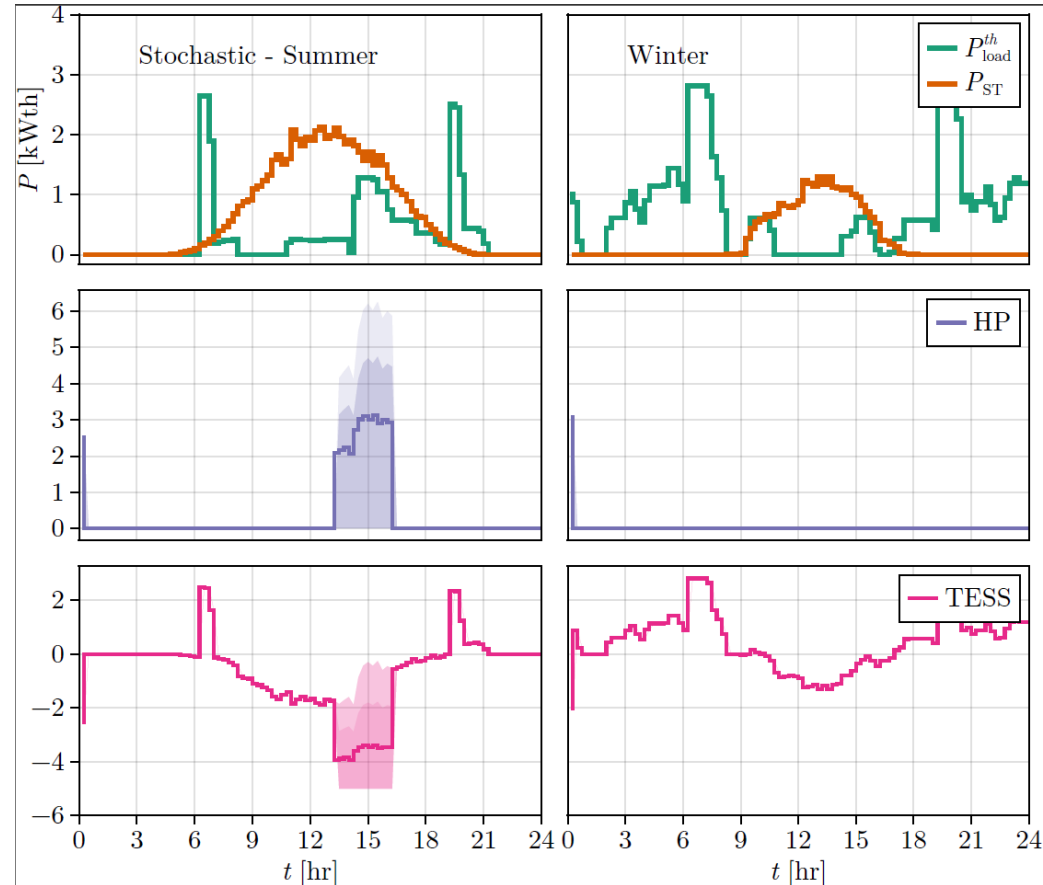
$\mathbb{E}_{t_{arr}}[C_i] + V_f$	Summer	Winter
RFO	-3,2	7,5
DET	0,2	8,2
Q_{BESS} [kWh]	Summer	Winter
RFO	50	48,4
DET	50	48,6



3. Case Study I

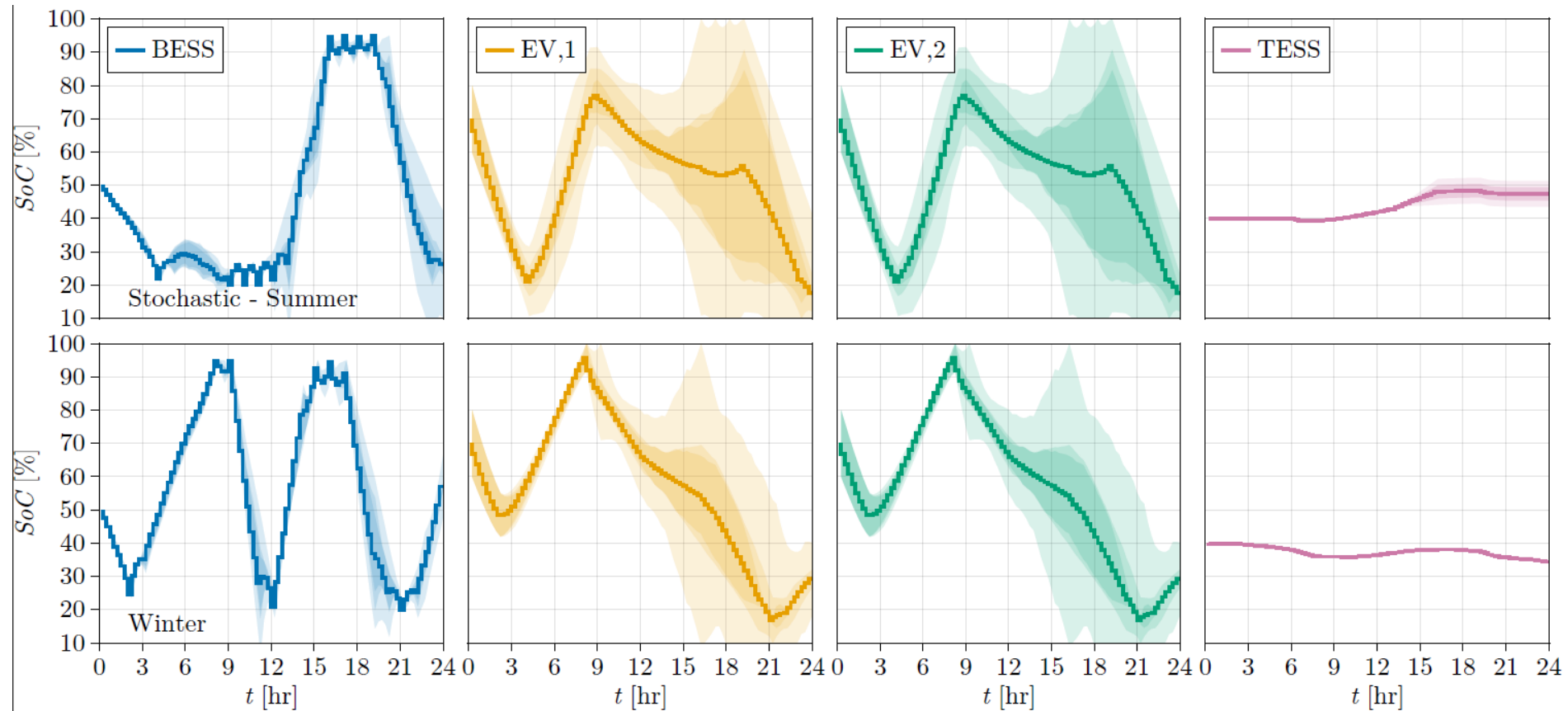
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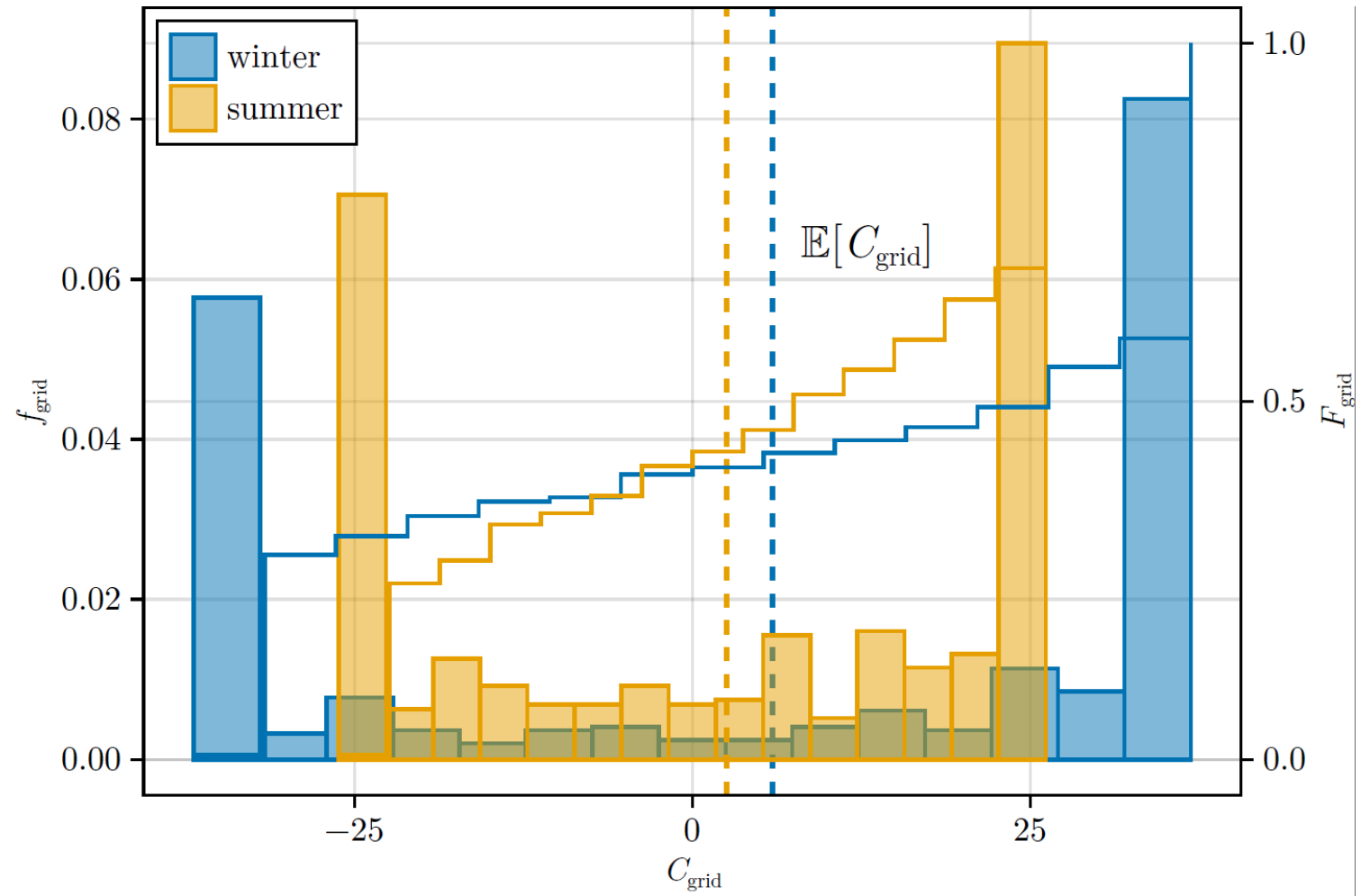
3. Case Study I

- SoC dispatch



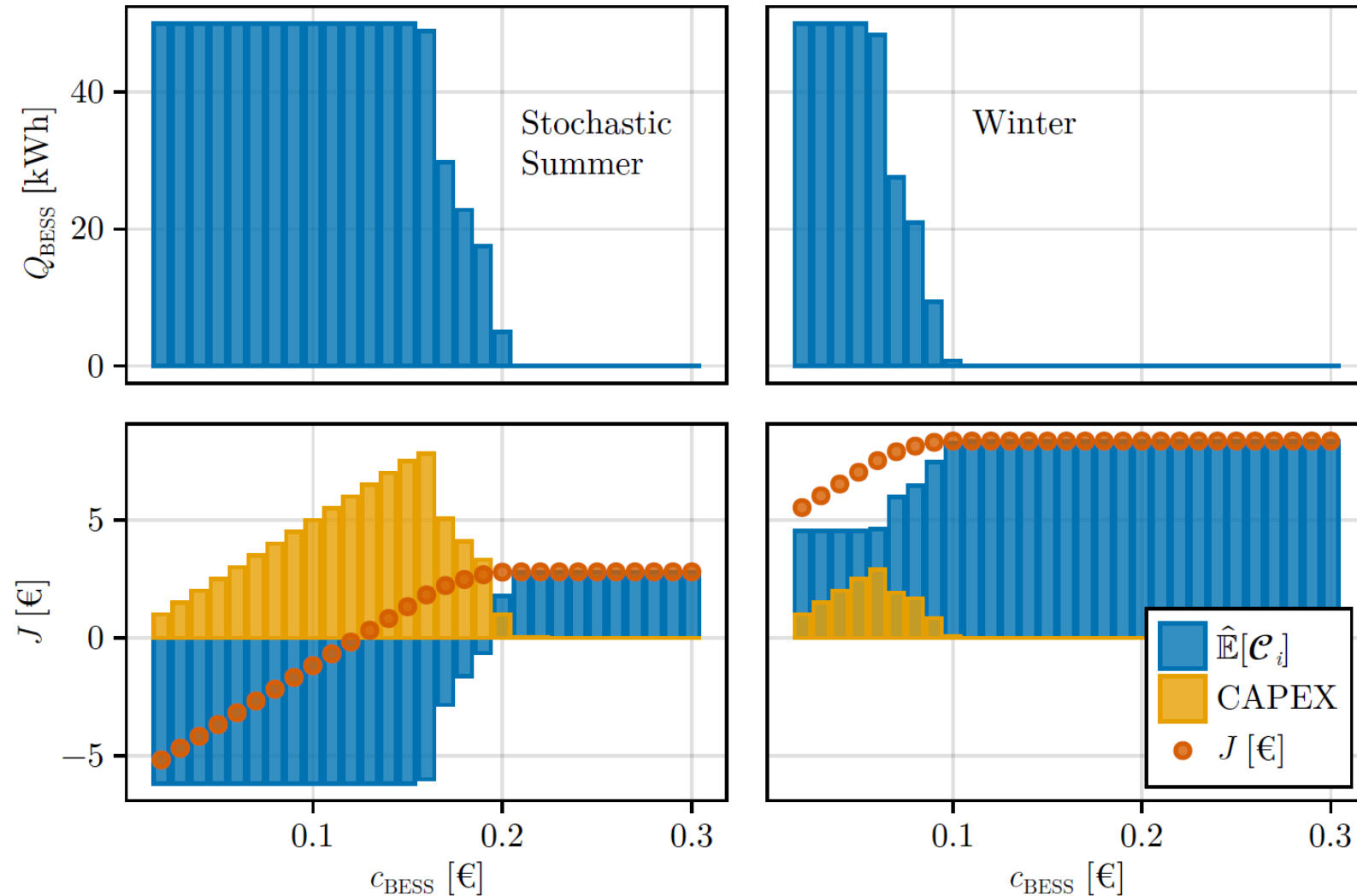
3. Case Study I

- Since the BESS capacity is the same for RFO and DET → The BESS does not support EV operations. → its capacity is maximized!
- It can provide other services!



3. Case Study II Sensitivity analysis

- Sensitivity against the BESS capacity.
- The breakpoint cost shifts with the season (the inputs)



4. Summary & Conclusions

- EV uncertainty is absorbed by the grid
- BESS can be used to supply/support other services

5. Ongoing research

- Extending analysis for other uncertainty sources and device capacities.
- Extending to multi-stage problems and long-term operation.



Github repo

He project is uitgevoerd met Topsector Energie subsidie van het Ministerie van Economische Zaken en Klimaat, uitgevoerd door Rijksdienst voor Ondernemend Nederland. De specifieke subsidie voor dit project betreft MOOI-subsidie ronde 2020



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Thank you!

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